Tabular Reinforcement Learning on Gymnasium Tasks

Reinforcement Learning, HW4–HW5 and Comprehensive Exercise Q1

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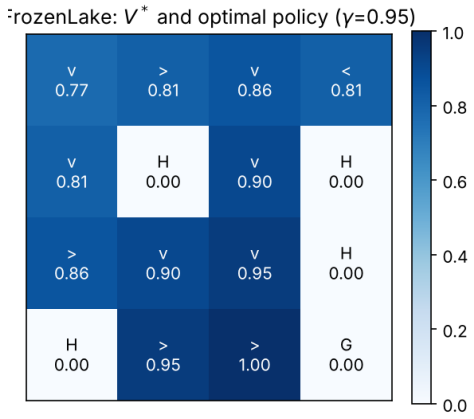
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1. FrozenLake: six solution methods

The deterministic 4×4 FrozenLake has 16 states and 4 actions; reaching the goal pays $+1$ and falling in a hole ends the episode. With $\gamma = 0.95$ I solved it with two model-based and four model-free methods.

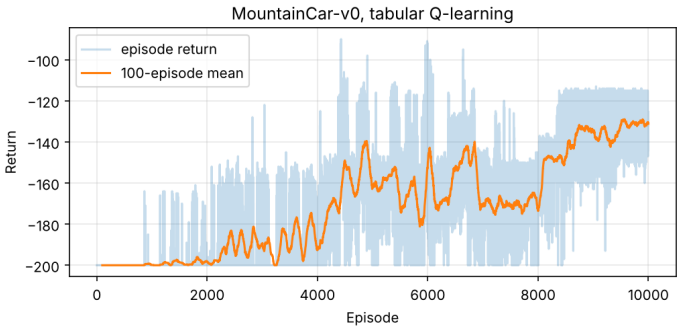
- **Policy iteration.** Exact evaluation $V^\pi = (I - \gamma P_\pi)^{-1} r_\pi$, then greedy improvement $\pi(s) \leftarrow \arg\max_a \sum_{s'} p(s'|s,a)\,[r + \gamma V^\pi(s')]$. Converged in 7 iterations.
- **Value iteration.** $V_{k+1}(s) = \max_a \sum_{s'} p(s'|s,a)\,[r + \gamma V_k(s')]$, 7 sweeps to 10^{-10}.
- **On-policy first-visit Monte Carlo** with an ε-soft policy, ε decaying from 0.5 to 0.05 over 20,000 episodes.
- **Off-policy Monte Carlo** with weighted importance sampling: greedy target policy, ε-greedy behaviour ($\varepsilon = 0.5$), $C(s,a)$ += W, Q += $\frac{W}{C}(G - Q)$, $W \leftarrow W/b(a|s)$, stopping at the first non-greedy action.
- **SARSA** and **Q-learning** with $\alpha = 0.5$ and ε decaying linearly to 0.01.

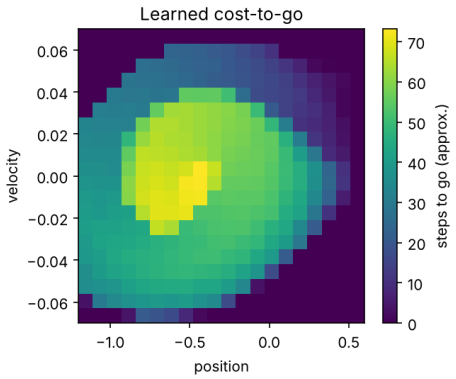
All six greedy policies reach the goal in 100% of rollouts. Policy and value iteration give the optimal action in all 11 non-terminal states; the model-free methods give it in 10–11, missing only states that the learned path rarely visits.



2. MountainCar: Q-learning on a grid

The continuous state (position, velocity) is binned on a 20×20 grid and Q-learning runs for 10,000 episodes with $\alpha = 0.1$, $\gamma = 0.99$ and ε decaying from 0.5 to 0.01. The greedy policy reaches the flag from all 100 test starts, in a median of 137 steps.





3. CartPole: Monte Carlo vs. Q-learning vs. SARSA

All three methods share one grid ($3 \times 3 \times 12 \times 12$ cells over cart position, cart velocity, pole angle and angular velocity), $\gamma = 0.99$, the same ε schedule and 6,000 episodes, and each ran with three seeds. The greedy policies balanced the pole for these mean numbers of steps over 50 test episodes (500 is the limit):

	Seed 0	Seed 1	Seed 2
Every-visit Monte Carlo	459.5	478.4	430.6
Q-learning	500.0	500.0	186.2
SARSA	500.0	257.8	296.0

The TD methods can reach the limit but depend strongly on the seed with a constant step size on a coarse grid. Monte Carlo, which averages complete returns, was the most consistent here.

